

Probability and Computing: Formulas

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$$\forall x, 1 + x \leq e^x$$

$$\forall x \in [0, 1], 1 - \frac{x}{2} \geq e^{-x}$$

$$\forall x \in (0, 1), e^{x/2} \leq 1 + x$$

$$\forall x \in (0, 1), \ln(1 + x) \geq \frac{x}{2}$$

$$\forall x \in (-1, 1), (1 - x)^y \approx e^{-yx}$$

$$\frac{e^x + e^{-x}}{2} \leq e^{x^2/2}$$

$$\left(\frac{n}{k}\right)^k \leq \binom{n}{k} \leq \left(\frac{en}{k}\right)^k$$

Stirling's formula

$$\forall m > 0, \sqrt{2\pi m} \left(\frac{m}{e}\right)^m \leq m! \leq 2\sqrt{2\pi m} \left(\frac{m}{e}\right)^m$$

Harmonic sum

$$H_n = \sum_{i=1}^n \frac{1}{i} = \ln n + \Theta(1)$$

Basel problem

$$\sum_{i=1}^n \frac{1}{i^2} < \sum_{i=1}^{\infty} \frac{1}{i^2} = \frac{\pi^2}{6} \implies \sum_{i=1}^n \frac{1}{i^2} = \Theta(1)$$

$$\forall x, y, -1 < x < y, \frac{x}{1+x} < \frac{y}{1+y}$$